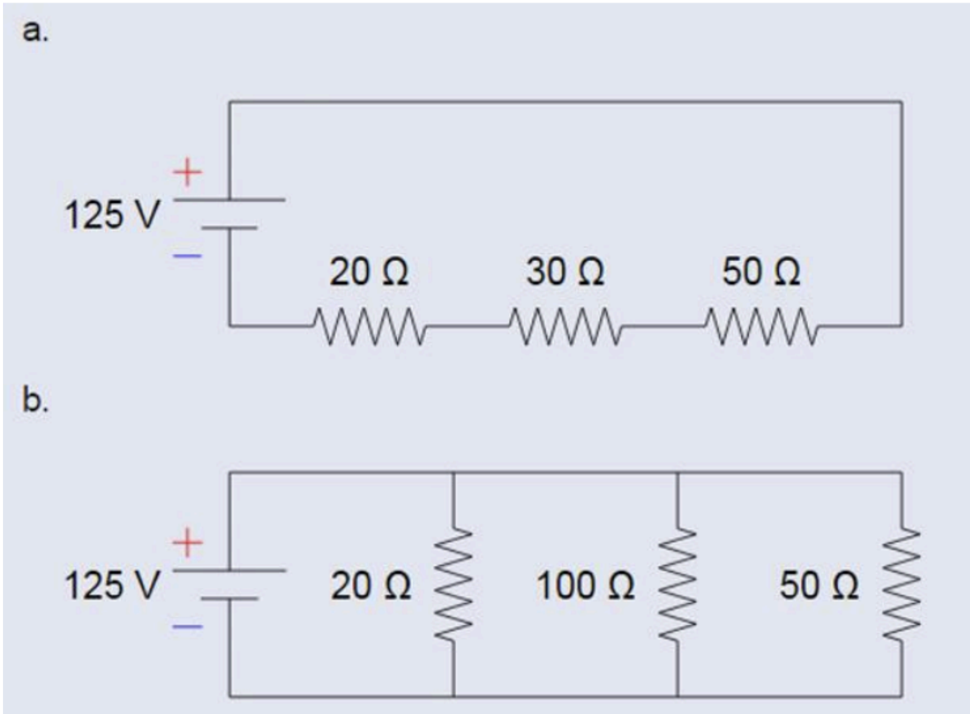
Eric Macharia Muthii(166390) - Circuit Calculations



1. Calculate the total resistance in ohms.

Circuit A:

Rtotal = 20 + 30 +50 = 100 ohms

Circuit B;

1/Rtotal = 1/20 + 1/100 + 1/50 = 0.08

1/Rtotal = 1/0.08

Rtotal = 12.5 ohms

2. Calculate the total current from the power supply

Circuit A;

I =V/R = 125/100 = 1.25 Amps

Circuit B;

I=V/R = 125/12.5 = 10 Amps

3. Calculate the current through each resistor

Circuit A;

Since the resistors are connected in series the current remains the same between all resistors which is **1.25 Amps**

Circuit B;
I20ohm = V/R = 125/20 = 6.25 Amps
I100ohm = V/R = 125/100 = 1.25 Amps
I50ohm = V/R = 125/50= 2.5 Amps

4.Calculate the Voltage drop through each resistor
Circuit A;
V20ohms = IR = 1.25*20 = 25 V
V30ohms = IR = 1.25*30 =37.5 V
V50ohms = IR = 1.25*50=62.5 V

Circuit B;
Since the resistors are in parallel the voltage drop will be equal to the source voltage which is 125 volts

5. Calculate the Power dissipated through each resistor (use Watt's Law: Power = Voltage * Current)
Circuit A;
P20ohms = VI = 25*1.25 =31.25 W
P30ohms = VI = 37.5*1.25 = 46.875 W
P50ohma = VI = 62.5*1.25 =78.125 W

Circuit B;
P20ohms = VI = 125*6.25=781.25W
P100ohms = VI = 125*1.25= 156.25W
P50ohms = VI = 125*2.5=312.5W